International Journal of Forecasting 42 (2026) 1163–1180

Contents lists available at ScienceDirect

International Journal of Forecasting

journal homepage: www.elsevier.com/locate/ijforecast

Machine learning and insurer failure prediction☆

Canchun He, Dejin Huang, Ruo Jia, Xi Wang*

School of Economics, Peking University, China

ARTICLE INFO

Article history:
Dataset link: https://github.com/CanchunHe/Machine-Learning-and-Insurer-Failure-Prediction

Keywords:
Insurer failure prediction
Machine learning
Artificial neural network
Insurance
Macroeconomic variables

ABSTRACT

Effective early identification of potentially fragile financial institutions is crucial for mitigating systemic risks due to the strong interconnection within the financial sector. For the first time, we apply the Structural Artificial Neural Network (SANN) to predict failure of financial institutions, using insurance companies as the context. The SANN is a category-wise machine learning algorithm (i.e., a category-aware learner) that extracts nonredundant information from Big Data with different categories of information. It enables the grouping of predictors into economic categories and thus allows interactions both within and across categories. We show that the SANN significantly improves the out-of-sample predictions of insurer failures, resulting in an average increase in the area under the curve (AUC) of 1.83% to 9.55% compared to traditional machine learning and logistic models, based on a sample of 2424 insurers from 17 European countries. We also document that macroeconomic and yield information provides nonredundant information in forecasting the failures, in addition to firm characteristics.

© 2025 International Institute of Forecasters. Published by Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1. Introduction

The solvency and sustainability of financial institutions play critical roles in the processes of the business cycle and financial turbulence. Insolvencies of financial institutions can trigger and amplify negative shocks, propagating systemic risk throughout the financial system. The 2008 financial crisis calls for further attention to the predictability and reliability of bankruptcy prediction models (Petropoulos et al., 2020; Veganzones et al., 2023). Detecting the risk of insolvency before its occurrence can mitigate systemic risks due to the strong interconnection among financial firms (Harrington, 2009; McDonald & Paulson, 2015). Therefore, effective early identification of potentially fragile financial institutions is crucial to enable regulators and practitioners to respond promptly and decisively (Eling & Jia, 2018; Zhang & Nielson, 2015).

This paper is the first to propose using a novel machine learning method, the Structural Artificial Neural Network (SANN), to predict the insolvencies and failures of financial institutions. Machine learning methods have been used widely in asset pricing (Gu et al., 2020) and insolvency prediction (Petropoulos et al., 2020). Different than traditional learning techniques, the advantage of the SANN lies in extracting and aggregating multiple types/categories of information. The categorization can be motivated by economically driven choices and characterize the network structure. Due to this category-wise structure, the SANN identifies the essential type of nonlinearity in predicting insolvency and failure. Specifically, the SANN detects whether the interaction between categories of information is more important than the nonlinear relationship within the same category.

To demonstrate the advantages of the SANN, we use a substantial sample consisting of 2424 insurers across 17 European markets to evaluate the out-of-sample performance of eight benchmark failure prediction models and two SANN models. The safety and stability of insurers are particularly important, because policyholders rely on

☆ The authors are grateful for helpful comments from the editors and two anonymous referees. The numerical results presented in this manuscript were reproduced by the Editor-in-Chief on 20 December 2025.

* Corresponding author.
E-mail addresses: HCC2018@pku.edu.cn (C. He), huangdejin@stu.pku.edu.cn (D. Huang), ruo.jia@pku.edu.cn (R. Jia), wang.x@pku.edu.cn (X. Wang).

https://doi.org/10.1016/j.ijforecast.2025.12.004
0169-2070/© 2025 International Institute of Forecasters. Published by Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

insurers to safeguard their lives and businesses and pay premiums for risk transfer and safety. The insolvencies of insurers also have strong connections to the systemic risk of the financial system. European markets particularly suit our purpose, given the diverse attitudes toward the tolerance of insurers' failures exhibited by their respective national regulators. These attitudes range from the rather lenient approach in the U.K. market to the highly conservative stance adopted in the German market.

Bank failure prediction has long been a central topic in forecasting research (Forgione & Migliardo, 2018; Gogas et al., 2018; Petropoulos et al., 2020). Previous studies typically set up a classification problem to predict the insolvency of banks; for example, Cox and Wang (2014) and Kočišová and Mišanková (2014) explored discriminant analysis, and Ohlson (1980) and Cole and Wu (2017) used Logit/Probit regression for classification. Along with the development of machine learning techniques, such methods as support vector machines, tree-based models, and neural network (NN)–type methods also have been explored by researchers (e.g., Chen & Shih, 2006; Gogas et al., 2018; Messai & Gallali, 2015). Recently, Petropoulos et al. (2020) used a horse-racing exercise to compare the performance of the aforementioned traditional machine learning methods. In this sense, our research supplements the banking insolvency studies with additional evidence from the insurance sector and a new method.

In this paper, we incorporate three categories of predictors to capture the full spectrum of factors that have potential impacts on insurer insolvency and failure. The existing literature considers incoming and outgoing cashflows (Chan et al., 2003; Feng, 2009, 2011; Konstantinides & Li, 2016), the environmental factors that affect all insurance lines of business (Delsing et al., 2019), and the return rates in the financial market that take financial risks into account (Chen & Yuan, 2017; Sun & Wei, 2014). These considerations provide a solid foundation for the classification of insolvency/failure predictors in this paper as (1) firm characteristics, which contain predictors covering accounting-based factors, for example, profitability, size, capital adequacy, investment, volatility, liquidity, company type, firm growth, reinsurance, and diversification (see Table 3); (2) macroeconomic indicators, which capture the variance of market-based factors in respective countries, including growth, market maturity, demographics, financial market, and general economic conditions, among others (see Table 4); and (3) yield curves (term structure), which refer to the 1-year to 10-year yields of sovereign debts in respective country-years (see Table 5), describing risk-free rates in the respective markets.

As benchmarks, we use the logistic classification method along with various conventional machine learning techniques, including decision trees, Random Forest (RF), XGBoost, and Artificial Neural Networks (ANNs). Different from traditional ANN models, our SANN approach adopts three separate ANNs to extract information from three different types of information: firm characteristics, macroeconomic indicators, and yield curves. Specifically, the SANN trains a category-wise network within each of the three categories and distills information from each category before further interaction. The information produced by each ANN independently serves as an input for prediction. Fundamentally, the SANN recognizes the relative importance of different types of nonlinearity and is able to detect whether the interactions between various categories of information or the nonlinear relationships within each category are more crucial in forecasting insurer insolvency and failure.

Our findings reveal that NN models outperform both the logit model and the decision tree models. Among these NN models, the SANN emerges as the superior choice due to its ability to extract nonredundant information from high-dimensional predictive variables. We find that both macroeconomic and yield variables contribute nonredundant predictive information, with macroeconomic variables offering predictive insights not captured by the yield curve alone. Furthermore, our results indicate that the nonlinearity within each predictor category is the key driver of the network's superior performance. Conversely, interactions after the distillation of information within predictor categories do not further enhance prediction accuracy.

Contribution and relation to the literature. Our paper contributes to the literature on insolvency and failure prediction of financial institutions. The existing empirical models for insolvency/failure prediction fall into two broad classes: statistical methods and intelligent techniques (Kumar & Ravi, 2007). Statistical methods include multivariate discriminant analysis (Meyer & Pifer, 1970; Sinkey, 1975), logistic regression (Campbell et al., 2008; Chen & Wong, 2004; Cheng & Weiss, 2012; Martin, 1977; Rauch & Wende, 2015; Zhang & Nielson, 2015), hazard models (Eling & Jia, 2018; Tian et al., 2015), and partial least squares (Mselmi et al., 2017). Statistical models depend on the assumptions of certain data distributions and thus may have relatively low accuracy when the assumptions are violated (Kumar & Ravi, 2007; Wang et al., 2014).

In recent years, artificial intelligence techniques, particularly machine learning, have become prevalent in the financial literature. AI techniques to predict insolvencies and failures include NNs (Brockett et al., 1994, 1997, 2006; Geng et al., 2015; Tam & Kiang, 1992), XGBoost (Borchert et al., 2023; Carmona et al., 2019; Climent et al., 2019), RF (Tanaka et al., 2016); (du Jardin, 2021), and support vector machines (SVMs) (Eling & Jia, 2018; Geng et al., 2015).

Several studies compare the performance of different machine learning methods. Lee and Urrutia (1996) show that the hazard model and the logit model have comparable forecasting accuracy. Brockett et al. (2006) show that NN methods outperform statistical approaches in predicting life insurers' financial distress. Carmona et al. (2019) find that XGBoost has greater predictive power than logistic regression and RF methods. Barboza et al. (2017) show that the RF method outperforms a series of benchmark approaches, including logistic regression and NN, for business bankruptcy prediction. Recently, Petropoulos et al. (2020) show that the RF has superior out-of-sample and out-of-time predictive performance, with NN performing almost equally well as RF in out-of-time samples.

In the era of Big Data, it is critical to explore prediction techniques that enable researchers to extract and aggregate predictive signals from high-dimensional information. Existing research shows that macroeconomic predictors, in addition to firm-specific variables, contribute to insolvency/failure prediction. Browne and Hoyt (1995) show that macroeconomic factors predict the failure of property-liability insurers. Zhang and Nielson (2015), Cheng and Weiss (2012), and Pasiouras and Gaganis (2013) incorporate both macroeconomic and firm-specific factors to confirm the important role of macroeconomic indicators. Doumpos et al. (2017) find that both country-level macroeconomic information and sector-level information contribute to prediction performance. However, macroeconomic and firm-specific factors are different types of data.

An ideal prediction technique should support category-wise analysis, so that information can be distilled from each category before further interaction. Building on this idea, Bianchi et al. (2021) introduced the SANN, which extracts and aggregates category-wise information. The idea of category-aware learning has been explored in the asset-pricing literature (e.g., Bianchi et al., 2021; Chen et al., 2023). To the best of our knowledge, to date no empirical work has applied a category-aware intelligent method to insolvency/failure prediction. **Our paper fills this gap by adopting the SANN to predict insolvency and failure of financial institutions. We show that the SANN outperforms the prevailing models in the literature.**

From a machine learning perspective, we make two additional contributions. First, we demonstrate that pooling heterogeneous data categories overloads conventional intelligent and statistical models. Risks of overfitting can conceal the nonredundant predictive information of macroeconomic indicators and yield curves. Second, an economic prior-designed network balances information extraction while reducing the risk of overfitting, achieving higher out-of-sample performance. Our work is the first to explore the difference between new machine learning and classical methods in predicting insurer insolvency/failure, which is a binary classification problem different from regression problems, such as in asset pricing (e.g., Gu et al., 2020).

The rest of the paper is structured as follows. Section 2 describes the dataset and our sample. Section 3 introduces our prediction methodology. Section 4 reports our empirical results. Section 5 checks the robustness of the results, and Section 6 concludes.

2. Data and sample

The dataset consists of three categories of information: (1) firm characteristics (firm financial information), (2) macroeconomic indicators, and (3) yield curves. A complete list of the 94 variables used in our analyses can be found in Tables 3–5.

Firm financial information was obtained from Best's Insurance Reports, which are widely used in the financial literature (e.g., Biener et al., 2017; Ge, 2022; Koijen & Yogo, 2022; Pasiouras & Gaganis, 2013). We used Europe-based insurers in the non-US Best's Insurance Reports (Best, 2002-2013) that are also operating insurance companies, excluding entities that are branches, special purpose vehicles, Lloyd's syndicates, captives, reinsurers, consolidated groups, and firms that operate insurance as a minor business, such as the banks, manufacturers, and health care providers identified by A.M. Best.

The dataset contains eight years of information, including failure predictors from 2005 to 2012 and corresponding failure events from 2006 to 2013, yielding 20,095 firm-year observations. We focus on countries with at least one firm failure event between 2006 and 2013. We drop observations containing missing values in firm-level scale-sensitive indicators,[1] macroeconomic variables, and/or yield curves. Thus, the final sample contains 2424 firms from 17 countries and 13,832 firm-year observations.

We obtain the failure events of insurers from the Best's Corporate Changes and Retirements database provided by Best (2002-2013). These events allow users to identify changes impacting existing companies, locate surviving insurers for companies that have gone out of business and more (Best, 2002-2013, 2010, 2013). The failure events in our sample include being in liquidation or liquidated (26.79%), in runoff (37.55%), license revoked (0.19%), merged (5.28%), struck from the register (0.57%), portfolio transfer (14.34%), inactive (14.53%), and insolvent (0.75%). Among these situations, being inactive or insolvent is defined as a failure event following the business status recorded in Best's Insurance Report. We construct a dummy variable *failure* to indicate whether a failure event occurs for the firm in the next year. At the end of year t, we predict whether firm i will have a failure event in year $t+1$. In Section 4.4, we report another dummy variable, *distressed*, constructed to indicate whether the firm experienced failure events in the next two years and to predict at the end of year t whether firm i will have a failure event in years $t+1$ and $t+2$. Tables 1 and 2 summarize the yearly distribution and country distribution of our sample, respectively.

The predictors of firm characteristics are classified into ten categories: profitability, size, capital adequacy, investment, volatility, liquidity, company type, firm growth, reinsurance, and diversification. We select firm-level predictors with missing values of less than 20% in the original dataset (i.e., 20,095 observations), yielding 55 indicators in total (see Table 3). In a robustness check, we analyze life and nonlife insurers separately and select firm-level predictors following the same standard (less than 20% missing values).[2]

To account for inflation, we adjust all monetary values by the CPI into real values in USD in 2013 (Biener et al., 2021; Eling & Jia, 2018). We take the natural logarithm

[1] These predictors are *net premiums written*, *gross premiums written*, *total assets*, *total liabilities*, *capital surplus*, *profit or loss before tax*, *profit or loss after tax*, *net technical reserves*, *gross technical reserves*, *total debtors*, *liquid assets*, *total investments*, *number of lines of business*, *number of premium regions*, and *number of employees*.

[2] We select firm-level predictors whose missing values are less than 20% in the life dataset and the nonlife dataset, respectively. The difference in predictors is described in Section 5 in detail.

Table 1
Yearly Distribution of Sample.

Year	Number of firms	Number of failure events	Failure rate[a]
2005	1354	/	/
2006	1822	37	2.73%
2007	1879	51	2.80%
2008	1932	86	4.58%
2009	1922	86	4.45%
2010	1708	101	5.25%
2011	1687	70	4.10%
2012	1528	60	3.56%
2013	/	39	2.55%
Total number of observations	13,832	530	3.83%

[a] Failure rate = Number of failure events$_{t+1}$/Number of firmst. The numerator in year t+1 refers to the total number of events that occur across all firms in year t.

Table 2
Country Distribution of Sample.

Country	Number of observations	Number of failure events	Failure rate
Austria	214	1	0.47%
Belgium	437	11	2.52%
Croatia	120	1	0.83%
Czech Republic	147	5	3.40%
Denmark	775	32	4.13%
Finland	236	5	2.12%
France	1213	12	0.99%
Germany	3181	13	0.41%
Ireland	741	103	13.90%
Italy	845	16	1.89%
Netherlands	924	21	2.27%
Portugal	224	1	0.45%
Slovakia	84	8	9.52%
Spain	1616	17	1.05%
Sweden	574	4	0.70%
Switzerland	785	13	1.66%
United Kingdom	1716	267	15.56%
Total	13,832	530	3.83%

of all firm-level scale-sensitive indicators.[3] We censor all firm characteristics (except dummies) at the 1st and 99th percentiles and then standardize them as the demeaned values divided by their standard deviations. We assign 0 to the missing values of scale-insensitive indicators after standardization (Green et al., 2017).

The macroeconomic indicators are obtained from Bloomberg, International Labor Organization (ILO), Thomson DataStream, and Wind databases. A total of 29 macroeconomic indicators are used at the country-year level, including 11 growth indicators, 4 market maturity indicators, 4 demographics indicators, 4 financial market indicators, 3 general economic conditions, and 3 others (see Table 4). We use annual growth rates of the non-ratio variables and standardize them as the demeaned value divided by its standard deviation (Ang & Piazzesi, 2003; Bai & Ng, 2008; Diebold et al., 2006). We also use macroeconomic variables in the natural logarithm of absolute values to capture the economic size and affluence of countries, including GDP, yearly wage, and GDP per capita. The yield curves obtained from Bloomberg capture the 1-year to 10-year yields of the sovereign debts in respective country-years (10 vars) (see Table 5). The three categories of information are combined according to the firm-domiciled countries and the years of the respective information.

[3] For indicators that have values smaller than 1 (i.e. *profit or loss before tax*, *profit or loss after tax*, *gross technical reserves*, *net technical reserves*, *total debtors*, *number of premium regions*, and *number of employees*), we add a fixed number (1 minus the minimum value of the variable) to all values of the variable to make sure its minimum value equal to 1 and then take the natural logarithm.

3. Prediction methodology

This section outlines our research design for the insurer insolvency and firm failure analysis. The first subsection describes the SANN methodology that we use with several benchmark methodologies. The second subsection briefly discusses the estimation and prediction strategy. The third subsection presents the criteria used to evaluate prediction performance.

3.1. Empirical methods

We take the logistic classification method as one of our benchmarks. Besides this statistical method, we adopt the RF, XGBoost, and ANN as conventional machine learning models. To better extract predictive information from different types of data (i.e., firm characteristics, macroeconomic series, and yield curves), we also adopt an aggregation algorithm of multineurons—the SANN—which we

Table 3
Firm characteristics.

Category	Variable	Mean	Min	Median	Max	Std. Dev.
Profitability	Profit or loss before tax	47,044.9	−136,098.2	4555.2	13,207,987	376,068.0
	Profit or loss after tax	38,871.6	−134,108.8	3464.9	13,289,487	366,506.7
	ROP	0.60	−22.3	0.054	135.5	7.49
	ROA	0.020	−0.40	0.012	0.37	0.058
	ROE	0.078	−2.78	0.083	1.44	0.24
	ROE before tax	0.10	−3.46	0.11	1.62	0.28
	ROA before tax	0.026	−0.46	0.015	0.40	0.065
	Moving average of ROA[1]	0.020	−0.39	0.012	0.33	0.043
	Moving average of ROP	0.43	−19.4	0.053	135.5	4.80
	Moving average of ROE	0.081	−2.38	0.080	1.17	0.17
	Moving average of ROA before tax	0.026	−0.42	0.015	0.40	0.051
	Moving average of ROE before tax	0.10	−2.73	0.10	1.32	0.20
	Risked-adjusted ROA[2]	190.9	0.0017	27.6	49,835.9	1261.3
	Risked-adjusted ROE	85.2	0.00036	38.7	27,684.1	539.6
	Risked-adjusted ROA before tax	150.0	0.0039	27.8	60,757.6	1091.2
	Risked-adjusted ROE before tax	160.4	0.00042	41.2	356,608.7	4658.6
Size	Gross premiums written	609,158.9	1.09	98,156.7	41,126,032	1,749,397.3
	Net premiums written	542,076.6	1.09	77,891.3	41,126,032	1,622,226.8
	Total assets	4,402,153.9	286.4	369,125.6	385,699.04	16,379,920.1
	Total liabilities	3,985,989.5	22.7	275,862.3	373,496.76	15,429,898.8
	Total debtors	170,524.7	0.067	15,020.2	16,680,624	681,116.0
	Gross technical reserves	3,617,682.2	0.86	227,940.0	348,554.12	14,305,971.8
	Net technical reserves	3,384,515.1	0.86	177,455.4	338,931.72	13,555,291.8
	Number of employees	1925.1	0.0021	228.7	2,349,352.5	20,999.5
Capital adequacy	Capital surplus	416,164.4	55.4	54,849.2	71,610,544	2,203,196.6
	Net premiums written/ capital surplus	2.45	0.000019	1.38	460.2	5.91
	Net technical reserves/ capital surplus	11.5	0.0000011	2.54	3116.1	38.9
	Gross premiums written/ capital surplus	2.83	0.000032	1.73	462.6	6.28
	Gross technical reserves/ capital surplus	12.4	0.0000011	3.26	3116.1	39.6
	Total debtors/ capital surplus	0.55	0.0000015	0.26	33.0	1.05
	Capital surplus/ total assets	0.27	0.00032	0.20	1.00	0.24
	Leverage ratio	13.5	0.0017	3.91	3125.6	40.9
Investment	Total debtors/ total assets	0.083	0.00000020	0.044	0.99	0.11
	Total investments	3,875,896.1	132.6	267,831.4	359,836.76	14,749,708.1
	Investment/ asset	0.80	0.0040	0.86	1.00	0.19
	Total investment/ total liabilities	1.95	0.0050	1.02	577.4	9.34
Volatility	Moving standard deviation of ROA	0.027	0	0.015	0.38	0.034
	Moving standard deviation of ROP	0.63	0	0.060	91.1	4.42
	Moving standard deviation of ROE	0.13	0	0.074	1.94	0.17
	Moving standard deviation of ROA before tax	0.030	0	0.018	0.36	0.036
	Moving standard deviation of ROE before tax	0.15	0	0.087	2.33	0.19
Liquidity	Liquid assets	3,050,910.1	5.57	207,142.9	347,746.56	11,963,283.5
	Liquid assets/ net technical reserves	22.5	0.0000062	1.13	81,900	878.2
	Liquid assets/ total liabilities	1.49	0.0000061	0.89	294.9	5.12
Company type	Mutual	0.20	0	0	1	0.40
	Composite	0.086	0	0	1	0.28
	Life	0.33	0	0	1	0.47
	Affiliated	0.66	0	1	1	0.47
Firm growth	Change in asset	0.095	−0.78	0.042	5.57	0.37
	Change in net premium	0.16	−0.94	0.015	22.5	1.23
	Change in gross premium	0.15	−0.93	0.013	22.3	1.21
Reinsurance	Premium retain ratio	0.84	0.000052	0.94	1.00	0.22
	Net technical reserves/ gross technical reserves	0.86	0.0044	0.98	1	0.21
Diversification	Number of lines of business	3.29	1	2	25	3.65
	Number of premium regions	0.35	0	0	7	0.62

Notes: 1. Moving average and moving standard deviation variables capture the long-term business results. We set the window width as five years (i.e., the year of the observation and the past four years). Within these five years, if the target variable has missing values, we use the non-missing values within the five years to calculate the moving variable. We only calculate values when the target variable has at least 3-year non-missing values within the five years. 2. Risk-adjusted variables are generated according to Biener et al. (2021).

further illustrate in what follows. This subsection briefly illustrates each model.

Decision trees and extensions (RF and XGBoost). A decision tree model is based on the partition of the input space into several subsets (i.e, leaves). For binary classification purposes, on each leaf we simply compute the probability of a positive event (e.g., an insurer failure). RF and XGBoost are extensions of decision trees. The RF method uses an ensemble of decision trees, and XGBoost takes the randomness of the partitions into consideration.

ANNs. The ANN model is an essential foundation for modern machine learning. The ANN mimics the behavior

Table 4
Macroeconomic indicators.

Category	Variable	Mean	Min	Median	Max	Std. Dev.
Growth	Real GDP growth	1.30	−8.54	1.69	10.6	3.05
	Change in yearly wage	0.045	−0.25	0.028	0.29	0.075
	Growth of industry production	0.36	−17.9	1.43	17.2	6.08
	Growth of capital	0.43	−19.7	1.52	16.9	7.40
	Growth of import	3.18	−20.3	5.45	19.2	7.51
	Growth of export	3.85	−20.1	5.39	22.8	7.46
	Growth of consumption per capita	0.63	−8.76	0.76	7.84	2.38
	Growth of gov expenditure	3.99	−25.3	3.73	36.7	5.16
	Growth of tax revenue	2.97	−15.7	4.14	14.2	5.78
	Unemployment rate	8.22	2.56	7.77	25.0	3.57
	Change in unemployment rate	0.036	−0.26	0.0056	0.88	0.19
Market maturity	Domestic private credit/ GDP	108.1	0	97.8	201.3	45.3
	Domestic credit/ GDP	147.2	41.6	147.3	248.9	54.3
	Insurance density	3191.7	278.3	2864.8	8598.8	2003.2
	Insurance penetration	0.076	0.028	0.076	0.19	0.032
Demographics	Change in population	0.0051	−0.020	0.0051	0.039	0.0059
	Labor force/ population	54.9	43.2	55.2	65.9	5.62
	Growth of population with access to Internet	0.061	−0.058	0.050	0.42	0.063
	Growth of labor in agriculture industry	−0.023	−0.21	−0.019	0.12	0.061
Financial market indicators	Inflation	2.15	−4.48	2.27	6.35	1.40
	Yearly return of MSCI	0.099	−0.054	0.100	0.34	0.078
	Change in IMF one year T-Bill rate	−0.022	−0.46	−0.043	0.63	0.14
	Change in exchange rate	−0.014	−0.97	−0.012	0.19	0.11
General economic conditions	GDP	916,864.8	44,821.1	403,188.7	3,619,161.5	1,002,848.1
	Yearly wage (insurance sector)	58,204.2	13,520.3	59,461.8	125,700	25,030.4
	GDP per capita	38,448.1	8844.4	40,634.6	82,583.6	15,068.0
Others	Change in current account	1.32	−2.28	0.044	163.5	14.0
	Capital ratio of banks	5.94	2.96	5.05	19.5	2.83
	Growth of carbon emission	−2.09	−18.1	−2.40	20.8	5.58

Table 5
Yield curves.

Maturity	Mean	Min	Median	Max	Std. Dev.
1Y	2.26	−0.27	2.04	12.8	1.86
2Y	2.53	−0.29	2.55	14.5	1.88
3Y	2.78	−0.23	2.81	16.2	1.95
4Y	2.97	−0.085	2.95	13.8	1.78
5Y	3.17	0.067	3.09	15.0	1.77
6Y	3.35	0.17	3.22	15.6	1.78
7Y	3.50	0.28	3.40	15.4	1.76
8Y	3.63	0.34	3.55	15.2	1.74
9Y	3.75	0.45	3.67	14.7	1.74
10Y	3.84	0.50	3.78	12.8	1.68

of biological systems composed of "neurons". With their powerful approximation ability,[4] ANN models are commonly used for pattern recognition in Computer Vision, Natural Language Processing, and Reinforcement Learning.

In our research context, we aim to identify the risky firms from the healthy ones. This is a typical binary classification problem, with two candidate categories: risky and healthy. Our true dataset is generated through $\Pr(\mathrm{y} = \text{risky}|\mathrm{x}) = f^*(x)$, mapping inputs x into probabilities. Through a parametric setting, the ANN defines a mapping $\Pr(\mathrm{y} = \text{risky}|\mathrm{x}) = f(x;\omega)$ and learns the parameters ω to approximate $f^*(x)$, the probability that a given firm with characteristics x turns out to be at risk.[5]

The ANN is also called a "feedforward network" since the connections between the nodes do not form cycles. Information evaluated from the inputs flows through the intermediate computation (hidden layers) and finally generates the output (prediction). Taking Fig. 1 as an example, we have inputs of 6 dimensions. Accordingly, the input layer (layer 0) has 6 neurons. Additionally, we have 2 hidden layers. The first hidden layer has 4 neurons, and the second has 2 neurons. For each layer except the input layer, all neurons in a layer take information from the previous layer. Finally, the output layer (with only one neuron) takes information from the last hidden layer to generate the final prediction. Therefore, an ANN can be written recursively. We explore information from firm characteristics, macroeconomic indicators, and yield curves simultaneously through the ANN, as each of them serves as an input, as shown in Fig. 1.

As benchmarks, we use several ANN models for our binary classification task.[6] We consider architectures with

[4] The goal of an ANN is to approximate some functions f^*. The well-known "Universal Approximation Theorem" (Funahashi, 1989; Hornik et al., 1989) for NNs states that under mild conditions, ANN is able to approximate any continuous function.

[5] There are two types of parameters. The first type is "model parameters", which are the coefficients of predictors learned from the training data, noted as ω, i.e., each $W^{(i)}$ and $b^{(i)}$ in layer-i, in Section 3. Model parameters are not manually set but are optimized through algorithms. The second type is "hyperparameters", which define the learning process, including for example the learning rate and the batch size. The optimal values of hyperparameters are pinned down through experimentation using the validation data.

[6] The ANN here is also known as a multilayer perceptron (MLP). In the context of deep learning, a perceptron is usually referred to as

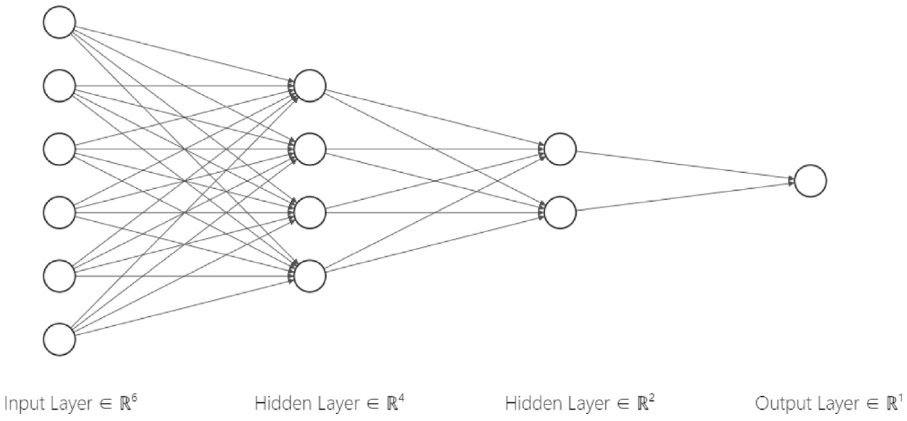


Fig. 1. An example of ANN.

Table 6
Illustration for Sample Partition.

	Prediction: Fail	Prediction: Survive
Facts: Fail	TP	FN
Facts: Survive	FP	TN

up to five hidden layers. Following Gu et al. (2020), our shallowest network has a single hidden layer with 4 neurons, which we denote by NN[4]. With similar notation, NN[4, 2] has two hidden layers with 4 and 2 neurons, respectively, and NN[4, 3, 2] has three hidden layers with 4, 3, and 2 neurons, respectively. We also evaluate deeper architectures including NN[8, 6, 4, 2] and NN[8, 6, 4, 3, 2]. This way of picking the number of neurons follows the pyramid rule, with the number of neurons gradually decreasing as the number of layers increases, creating a pyramid-like structure (Masters, 1993).

An ANN processes information through a sequence of nonlinear operations. Formally, let $x^{(i)}$ denote the information that we get from layer i, where $x^{(i)}$ is a vector with dimension $k^{(i)} \times 1$. While $i = 0$ represents the input layer, $i > 0$ denotes the hidden layers or the output layer. The information $x^{(i+1)}$ produced by the layer $i + 1$ is

$$x^{(i+1)} = h\left(W^{(i)}x^{(i)} + b^{(i)}\right), \tag{1}$$

where $W^{(i)}$ is a matrix with dimension $k^{(i+1)} \times k^{(i)}$ and $b^{(i)}$ is a vector with dimension $k^{(i+1)} \times 1$. The function $h(.)$ is a nonlinear activation function that transforms each element of the vector nonlinearly. For the hidden layers, we adopt the rectified linear unit (ReLU) activation function, defined as $h(x) = \max(0, x)$. For the output layer, we adopt the sigmoid activation function, defined as $h(x) = \frac{1}{1+\exp(-x)}$, which is particularly well suited for binary classification tasks. With an increasing number of hidden layers, we enter the region called "deep learning".

SANNs. We are investigating not only whether machine learning methods defeat traditional methods, but also how economic intuition helps to discipline the machine learning models. This approach takes advantage of both economic intuition-based guidance and data-driven machine learning methods and thus is expected to achieve better performance.

Generally, we consider firm characteristics, macro variables, and the yield curve to approximate the information of firm operation (micro level), macroeconomics (macro level), and the financial market environment. Intuitively, we would like to use some methods to aggregate the information separately, extract the predictive information, and explore their predictive power.

Different from traditional ANN models, we use an SANN created by Bianchi et al. (2021) to extract and aggregate different types/categories of information. We illustrate the construction of the SANN in Fig. 2. As building blocks for the SANN, we design a nonoverlapping ANN for each type of information. Fig. 2 illustrates the difference between the SANN and the ANN; although the SANN extracts nonlinear information from each type/category of inputs, there is no further interaction between each type of information in SANN0 (left panel of Fig. 2). There are no further hidden layers following category-wise ANNs. Surely SANN0 can be extended to SANN1 to incorporate interactions between different types of distilled information, as shown in the right panel of Fig. 2.

As shown in Fig. 2, we use three ANNs to extract information from firm characteristics, macro variables, and the yield curve separately. Each ANN produces information that independently serves as input for the final prediction. In other words, the final layer of each (pre-)ANN model serves as a neuron in the final hidden layer in our framework.

Therefore, the SANN can be viewed as an economically driven choice of network structure. Grouping variables within the same economic category and then training a category-wise network within each group can distill information from each category before further interaction. According to the universal approximation theorem, NNs with more parameters (higher complexity, e.g., deep NNs) are able to extract information for lower prediction bias than NNs with less complexity (e.g., shallow NNs). However, higher-complexity NNs prone to overfitting, especially when dealing with high-dimensional

a neuron; an MLP refers to a feedforward ANN consisting of multiple layers of interconnected neurons.

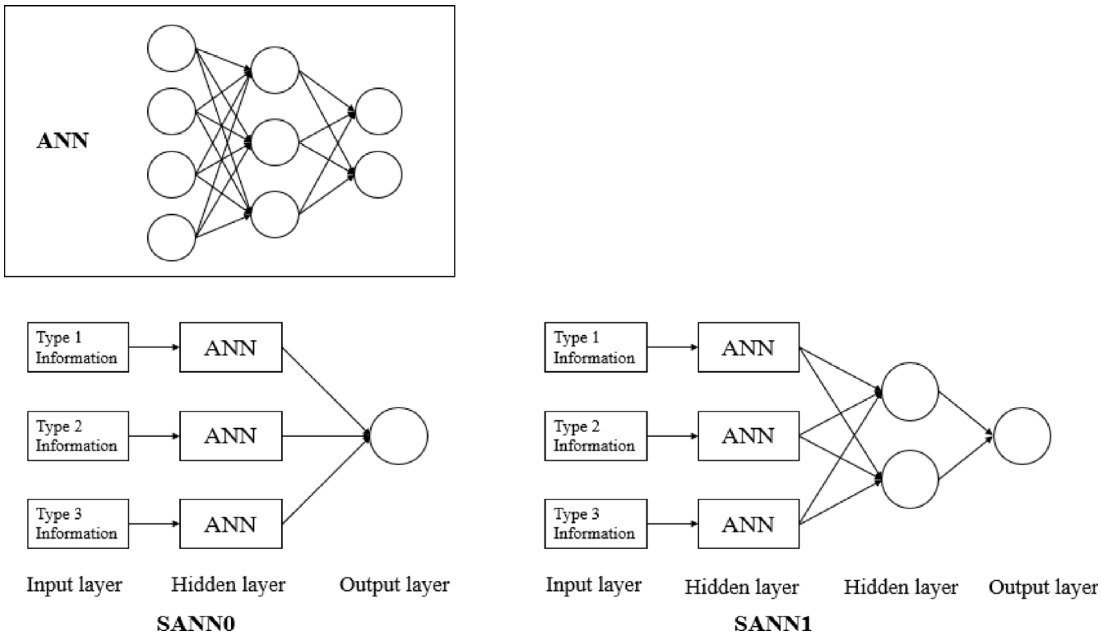


Fig. 2. An illustration of SANN.

Table 7
Out-of-sample AUC in Detecting Firm Failures.

Model	logit	RF	XGBoost	NN[4]	NN[4,2]	NN[4,3,2]	NN[8,6,4,2]	NN[8,6,4,3,2]	SANN0	SANN1
Information Explored										
Firm	0.8481	0.8422	0.8357	0.8607	0.8644	0.8538	0.8613	0.8161	0.8538	0.8538
Firm and yield	0.7831	0.7982	0.7660	0.8540	0.8201	0.8273	0.8580	0.8506	0.8613	0.8651
Firm and macro	0.8118	0.7775	0.8021	0.8496	0.8449	0.8399	0.8638	0.8382	0.8682	0.8745
All	0.7860	0.8109	0.7623	0.8368	0.8295	0.8313	0.8233	0.8441	0.8853	0.8824

data, echoing the variance–bias tradeoff. In our context, take Table 7 for instance, which shows that as the NN becomes more complex, the out-of-sample AUC generally increases (from NN[4, 2], NN[4, 3, 2] to NN[8, 6, 4, 2]) but declines from NN[8, 6, 4, 2] to NN[8, 6, 4, 3, 2]. To address this issue, the traditional literature usually adopts cross-validation for hyperparameter tuning to find a model with proper complexity.

In contrast, the SANN adopts an economic prior–driven category-wise structure that balances information extraction, reducing the risk of overfitting. If the SANN outperforms the deeper NNs that do not group variables, we can infer that the economic discipline adopted by the SANN can provide another routine to address the variance–bias tradeoff. To the best of our knowledge, these specifications have never been applied in the empirical finance literature in the context of predictive classification methods.

Furthermore, the SANN, especially the SANN0, can determine what type of nonlinearity is important in predicting insurer insolvency and failure and also whether the interaction between types of information or the nonlinear relationship within the same type of information is more important. The asset pricing literature exploits the importance of nonlinearity between predictors in predicting stock returns (Chen et al., 2023; Gu et al., 2020) and identifies the relative importance of different types of interactions (Bianchi et al., 2021; Chen et al., 2023). This is the first paper to characterize the relevant interaction in predicting insurer insolvency and failure.

As shown in Fig. 2, to extract information from firm, macro, and yield data, our networks for each single type of information have three hidden layers with 4, 3, and 2 neurons. The SANN0 model based on firm and macro (yield) data has two separate ANNs modules to extract information, while the SANN0 model based on all data has three separate ANNs to extract information. The only difference between the SANN1 and the SANN0 based on the same data is that the SANN1 has an extra hidden layer with 2 neurons after the separated ANN modules to allow further interaction between different data types. Therefore, we refer to the SANN1 as a postinteraction shallow category-wise network.

In sum, we use 10 models in our research: logit, RF, XGBoost, NN[4], NN[4,2], NN[4,3,2], NN[8,6,4,2], NN[8,6,4,3,2], SANN0, and SANN1. By comparing the performance of each model under different information sets, we show how SANNs dominate other prediction methods in insurer failure prediction.

3.2. Estimation strategy

Following the standard practice in machine learning, we divide the sample into a training sample, a validation

sample, and a test sample (Bianchi et al., 2021; Gu et al., 2020). We use the training sample for model estimation, the validation sample to pin down the hyperparameters of the models, and the test sample to make out-of-sample predictions and evaluate the model's out-of-sample performance.

We roll our algorithm forward for predictions. We make three out-of-sample predictions: firm failure in 2011 based on information in 2010, firm failure in 2012 based on information in 2011, and firm failure in 2013 based on information in 2012. When predicting failure in year t, the test sample we use is the predictors in year t - 1 and the corresponding failure events in year t. The validation sample is the 1-year data before the year of the test sample, and the training sample is the data from all years before the year of the validation sample. For example, to predict firm failure in 2011, we use the predictors in 2005–2008 and the failure events in 2006–2009 for training, use the predictors in 2009 and failure events in 2010 for validation, and use the predictors in 2010 and the failure events in 2011 for out-of-sample prediction and evaluation.

We estimate the model parameters as follows. Given the potential hyperparameters, following the machine learning literature, we obtain the estimated parameters—i.e, each $W^{(i)}$ and $b^{(i)}$ in layer I, $x^{(i+1)} = h\left(W^{(i)}x^{(i)} + b^{(i)}\right)$ – through optimization of an objective function. For binary classification purposes, our objective function (or minus loss function) is defined as an empirical log-likelihood function (the corresponding loss function is also called cross entropy in the machine learning literature[7]),

$$J(\omega) = \sum_{n=1}^{N}[1(y_n = 1)\log f(x_n;\omega) + 1(y_n = 0)\log(1 - f(x_n;\omega))],$$

where N represents the number of observations in the training process and $f(x_n;\omega)$ denotes the predicted value of y_n, i.e., the prediction probability that y_n equals 1. Here $J(\omega)$ can be viewed as the cross-entropy between predicted probabilities and the realized probability and can be used to quantify how closely the model's predictions align with the true values.

We estimate the parameters by maximizing the objective function, $\omega^* = argmax_\omega J(\omega)$. Because of the high degree of nonlinearity and nonconvexity in models using NNs, the research community often adopts numerical algorithms (e.g., stochastic gradient descent or its extensions) to train an NN (Gu et al., 2020). In addition, we use "early stop" and "dropout" to regularize our estimation process.[8] Furthermore, to enhance the robustness of the optimization results, we use an ensemble approach by using 10 NNs—each initialized with a different random seed—to estimate every NN-type model. We generate 10 separate predictions from these networks and compute the final predicted value for each observation by averaging the 10 separate predictions.[9]

By comparing the values of the objective function across various potential hyperparameters in the validation sample, we pin down the optimal hyperparameter values. Table A.3 in Appendix describes the set of hyperparameters and their potential values used for tuning each machine learning method.

3.3. Prediction performance

We adopt the AUC criteria following the classical failure prediction literature (Bauer & Agarwal, 2014; Borchert et al., 2023; Chava & Jarrow, 2004). The AUC is less sensitive to data imbalance than other predictive accuracy measures (Wang et al., 2020). We follow Ting (2002) to adopt a "cost-sensitive learning" approach to achieve equal weights for failure and survival samples.[10]

For our binary prediction problem, we separate the sample into 4 categories: (1) true positive (TP), predicted to fail and actually failed in the specific year; (2) false positive (FP), predicted to fail but did not fail in the specific year; (3) true negative (TN), predicted to survive and actually survived in the specific year; and (4) false negative (FN), predicted to survive but failed in the specific year. We summarize this partition in Table 6.

We define the TP rate (TPR, i.e., recall rate) and the FP rate (FPR) as follows:

$$TPR = \frac{TP}{TP + FN}$$

and

$$FPR = \frac{FP}{FP + TN}.$$

As one can tell from definitions of the *TPR* and *FPR*, the *FPR* measures the type I error in classification (prediction), and $1 - TPR$ measures the type II error in classification (prediction). A type I error occurs when a surviving firm is incorrectly classified as failed, and a type II error occurs

[7] For details on the cross entropy, see Ho & Wookey, 2019, and Eelbode et al., 2020.

[8] "Early stop" is a technique used to halt the training process to prevent overfitting. In our prediction model, we define the stopping criterion such that if the objective function does not improve for 50 consecutive iterations, training is terminated. Similarly, "dropout" is a regularization method that randomly removes connections in NNs with a certain probability to mitigate overfitting. In our case, we set this probability to 10%.

[9] To show the robustness of the ensembled 10 NNs, we also conduct the following test. Within the 10 NNs, we perform 10 rounds of sampling, each time selecting 9 NNs, and each NN is left out once in one of the rounds. After each round of sampling, the average predicted value of the selected 9 models is used as the final prediction, and the AUC is calculated. Thus, for each NN model under each information set, one can also obtain 10 AUC values. In Table A.1 in Appendix, we report the standard deviation of the 10 AUCs. Since all standard deviations are very small (not exceeding 0.01), the 10 NNs initialized with 10 different random seeds are robust.

[10] We achieve equal weights to address the problem that machine learning algorithms tend to learn primarily from the samples of the majority class (survival sample) and overlook the unique characteristics and patterns in the minority class (failed sample) in the case of data imbalance. We perform an analysis without "cost-sensitive learning" and present the results in Table A.2 in Appendix. Our findings indicate that while our baseline conclusions remain robust, the predictive power decreases when incorporating all three types of information. This suggests that "cost-sensitive learning" enhances overall performance.

when a failed firm is incorrectly classified as surviving. There is a tradeoff between type I and type II errors. Setting a lower level of type I error implicitly increases the threshold to classify a sample as positive, which will lower the TPR and increase the type II error.

For any decision threshold θ, a model will classify a given sample as positive (failure) if the predicted probability of failure is larger than the threshold θ, i.e. $f\left(x_{i,t-1}, m_{t-1}, r_{t-1}\right) > \theta$. Then we can compute the TPR and FPR for a given threshold θ. Changing θ delivers a series of TPR and FPR pairs. A receiver operating characteristic (ROC) curve plots these (FPR, TPR) pairs on a coordinate system, illustrating the diagnostic ability of binary classifiers.

We use the AUC as a prediction performance measure. The AUC is defined as the area under the ROC curve (see the equation below). Because both the TPR and the FPR range from 0 to 1, the AUC is also within this range, with a higher AUC reflecting better performance. According to this definition, the AUC is a threshold-independent metric and accounts for the performance under all the possible threshold values (De Caigny et al., 2018):

$$AUC = \int_0^1 \frac{TP}{TP + FN} d\left(\frac{FP}{FP + TN}\right)$$

We compute the AUC for a specific model as follows. Given a model, we use the prediction process illustrated in Section 3.2 to produce predicted values for each observation in the test sample. By changing the threshold θ from 0 to 1, we get a sequence of distinct (FPR, TPR) pairs. With these (FPR, TPR) pairs, we plot the ROC curve and compute the area under the ROC curve (i.e., the AUC) accordingly. Considering the discrete (FPR, TPR) pairs on the ROC curve, we calculate the AUC using the trapezoidal rule, defined in the following equation:

$$AUC = \sum_{i=1}^{n-1} \frac{(x_{i+1} - x_i) \cdot (y_{i+1} + y_i)}{2},$$

where x_i is the FPR, y_i is the TPR, and n is the number of distinct (FPR, TPR) points on the ROC curve. This method calculates the AUC by dividing it into multiple trapezoids and summing their respective areas (Pedregosa et al., 2011).

4. Empirical results

4.1. Performance of SANN models in predicting failures

In Table 7, we report the out-of-sample AUC for various prediction methods. The logit achieves an out-of-sample AUC of 0.8481 with firm information only. As shown in column 1 of Table 7, although the logit model can achieve a decent classification performance, incorporating high-dimensional information is difficult. Comparing the performance of the logit model based on firm information and the logit model based on all firm, macro, and yield information, one may conclude that the macro and yield data provide noisy or at least redundant information for insurer insolvency and failure prediction. Adding macro data to firm information decreases the AUC of the logit model from 0.8481 to 0.8118, and further adding yield information decreases the AUC to 0.7860.

Table 7 also shows that NN models outperform the logit and the decision tree models (RF and XGBoost) when high-dimensional information is provided. Among all the machine learning methods, the decision tree models (RF and XGBoost) achieve similar performance to the logit model. The AUC of the RF exploring all information achieves 0.8109, the XGBoost exploring all information achieves 0.7623, and the logit model achieves 0.7860. NN models (including ANNs and SANNs) on average improve the AUC by 7.82% compared with the logit model when exploring all information.

Traditional NN models cannot address the curse of high dimensionality, however. Although NNs dominate the logit model in incorporating high-dimensional information, adding macro and/or yield information to the NN models decreases their own performance. Taking the NN [4,3,2] model as an example, adding macro information to the model based on firm information decreases the AUC from 0.8538 to 0.8399, and adding yield information decreases the AUC further to 0.8313. Moreover, increasing the number of NN layers does not continuously improve model prediction performance. The performance of 4-layer and 5-layer NN models is similar to that of 3-layer NN models.

Compared with models based on firm information only, the other two categories of information decrease the AUC by 7.32% for the logit model and by 2.64% for the NN[4,3,2] model. Although the decreasing speed of the NN is lower than that of the logit, the NN models as well as other traditional machine learning techniques are not able to extract nonredundant predictive information from macro and yield data.

The power of economic prior: Macro and yield information. Is yield or macro information redundant or even harmful in predicting insurer insolvency and failure? Is there any difficulty with the existing methods in processing the information? The answer is negative for the first question and positive for the second. The last two columns of Table 7 present the performance of the SANN0 and the SANN1. As demonstrated in Table 7, the SANN0 outperforms traditional NN models in processing macro and yield data. This superior performance shows that the SANN0 is able to capture nonredundant information from high-dimensional predictors. The SANN0 model extracts information from different categories of data separately, as shown in Fig. 2. If one does not differentiate between different categories of firm information, the SANN0 based on firm variables degenerates into the NN[4,3,2] model; however, as one adds more categories of information, the difference between the SANN0 and NN models becomes apparent. Based on firm and yield information, the SANN0 improves the 0.8273 AUC of the NN[4,3,2] to 0.8613, while based on firm and macro information, the SANN0 improves the 0.8399 AUC of the NN[4,3,2] to 0.8682.

Compared with the 0.8538 AUC of the SANN0 based on firm information only, adding yield information to the SANN0 increases the AUC by 0.88%, and adding both yield and macro increases the AUC by 3.70%. In other words, yield and macro data provide nonredundant insights,

Table 8
Out-of-sample False Positive Rate (Type-I Error Rate) in Detecting Firm Failures.

Model	logit	RF	XGBoost	NN[4]	NN[4,2]	NN[4,3,2]	NN[8,6,4,2]	NN[8,6,4,3,2]	SANN0	SANN1
Panel A: Type-II Error Rate = 10% (True Positive Rate = 90%)										
Firm	43.46%	46.23%	52.90%	40.03%	41.42%	46.38%	38.30%	45.44%	46.38%	46.38%
Firm and yield	55.66%	52.50%	53.32%	40.49%	40.64%	44.03%	41.90%	44.74%	38.89%	42.45%
Firm and macro	47.85%	55.07%	50.84%	39.97%	41.21%	40.85%	38.51%	52.71%	38.43%	36.03%
All	67.80%	43.46%	61.25%	47.43%	46.13%	45.39%	50.48%	40.30%	34.75%	35.09%
Panel B: Type-II Error Rate = 20% (True Positive Rate = 80%)										
Firm	25.14%	26.69%	28.36%	23.79%	21.06%	23.56%	22.40%	25.83%	23.56%	23.56%
Firm and yield	34.22%	35.34%	42.26%	24.72%	29.53%	26.84%	22.42%	24.91%	23.33%	23.81%
Firm and macro	33.82%	38.68%	37.86%	25.58%	24.86%	25.75%	23.41%	25.31%	24.27%	20.28%
All	37.08%	31.91%	45.37%	28.44%	28.61%	27.05%	30.10%	26.74%	18.99%	19.88%
Panel C: Type-II Error Rate = 30% (True Positive Rate = 70%)										
Firm	16.58%	18.87%	16.74%	16.64%	13.46%	14.81%	13.57%	14.09%	14.81%	14.81%
Firm and yield	24.38%	21.52%	31.83%	15.82%	21.14%	16.87%	15.63%	16.62%	15.52%	13.86%
Firm and macro	25.18%	30.65%	28.31%	18.47%	19.10%	17.08%	16.76%	16.34%	12.58%	10.56%
All	21.37%	24.06%	35.13%	16.91%	20.09%	17.19%	14.81%	17.02%	10.75%	11.25%

Note: We obtain the false positive rate (FPR) and true positive rate (TPR) by adjusting the threshold θ defined in Section 3.3. The θ values used to construct the ROC curve range from 1 to all predicted values yielding distinct (FPR, TPR) pairs. To report FPR at a fixed TPR, we identify the θ at which the TPR is closest to the fixed level and report the corresponding FPR. Since the θ values used for ROC construction are sufficiently dense, we can always find a TPR that closely approximates the fixed level, with a deviation of less than 0.5%. Consequently, the FPR values presented in the table can be regarded as the FPR at the specified TPR levels.

confirming the need to consider environmental factors and return rates in insolvency prediction (Chen & Yuan, 2017; Delsing et al., 2019; Sun & Wei, 2014).

Moreover, the SANN0 significantly improves out-of-sample failure prediction compared with the logistic and traditional machine learning models, with an average AUC improvement from the SANN0 considering the four information sets (i.e., firm, firm and yield, firm and macro, and all) ranging between 1.83% and 9.55%.

The SANN models defeat the best-performing deep NN models (i.e., the ANN models), which generally achieve the best out-of-sample AUC by exploring firm information only. In this sense, complementary to the NN's universal approximation ability, economic priors (i.e., macro and yield data in our case) enhance insurer failure prediction with nonredundant insights. In addition, macroeconomic variables include predictive information not spanned by the yield curve.

In Table 8, following Eling and Jia (2018), we report the FPR for each model, given the TPR levels of 90%, 80%, and 70%, respectively, under different information sets. As discussed in Section 3.3, the FPR is the type I error rate and 1-TPR is the type II error rate. We observe a decreasing trend of type I error rate when the fixed type II error rate increases, suggesting a trade-off between type I and type II errors. Given a fixed type II error rate, a lower type I error rate indicates better prediction performance. The results in Table 8 further confirm our baseline findings presented in Table 7.

4.2. Interaction across or within categories?

As shown in Fig. 2, the SANN0 completely ignores any interaction across different categories of information, while the SANN1 explores an extra hidden layer for aggregating information and allows for further interaction. To better illustrate the potential interactions of different information categories, we estimate the SANN1 model. Comparing the performance between the SANN0 and the SANN1, one can observe that the SANN1 outperforms the SANN0 only marginally in out-of-sample prediction. The AUCs of the SANN1 based on firm information only, on firm and yield information, on firm and macro information, and on all information are comparable to those of the SANN0 (with changes of 0.00%, 0.44%, 0.72%, and −0.33%, respectively). This suggests that the nonlinear relationships across categories contribute little to predicting insurer failure, aligning with the findings of Bianchi et al. (2021) in asset pricing.

Is the interaction across different categories of information or the interaction within each category more important? The slight improvement from allowing further interaction between categories implies that interactions within each category are much more critical than interactions across categories.

In sum, our results from comparing the SANN0 and the ANN show that grouping variables and distilling information can provide another routine for extracting predictive information. In addition, because the SANN0 shuts down the interactions between different categories of variables, the better performance of the SANN0 shows that nonlinearity within a group drives the outperformance of the network. Moreover, nonlinear interactions across information categories may be redundant in predicting firm failure; comparing the performance of the SANN0 and the SANN1 reveals that interactions after information distilling cannot further improve the prediction performance.

Our findings are novel to the literature on both machine learning and insurer insolvency and failure prediction in that instead of merely relying on the "universal approximation theorem" naively and bearing the potential cost of wiping out nonredundant predictive information, this paper uses economic theory as a complement to machine learning techniques in predicting firm

failure. In addition, the postinteraction shallow category-wise network (SANN1) does not outperform the shallow category-wise network without any interactions (SANN0), and a shallow category-wise network (SANN0 or SANN1) outperforms a deep NN that models firm characteristics, macroeconomics, and yield curves separately or together.

4.3. Relative importance of predictors

In this subsection, we investigate the relative importance of variables in predicting insurer insolvency and failure. To this end, we examine the marginal relevance of a single variable based on the partial derivative of the predicted failure probability with respect to each predictor, where the gradient is evaluated at the in-sample mean value of the predictor, that is,

$$E\left[\frac{\partial}{\partial x_{i,t}} f\left(x;\theta\right)|x_{i,t}=\bar{x}_i\right], \tag{2}$$

where $\bar{x}_i$ represents the in-sample mean of input variable i. The partial derivative represents the sensitivity of the output to the ith input evaluated at its sample mean, conditional on the network structure and the average value of all other input variables. We take the absolute value and the square of this partial derivative to evaluate the relative importance (Bianchi et al., 2021; Gu et al., 2020).

Fig. 3 summarizes the relative importance of each information category, and Figs. 4 to 6 summarize the relative importance of input variables within each category based on Eq. (2). This analysis is based on the SANN0 with all firm, macro, and yield information.

As shown in Fig. 3, firm variables provide the greatest predictive power, accounting for 63.09%. Macro variables are the second most important information source, contributing 29.46%. The yield curve is the least important category. This ranking is unsurprising, as the SANN0 based on firm and macro outperforms the SANN0 based on firm and yield, as shown in Table 7.

As shown in Fig. 4, *mutual* (−), *composite* (−), and *moving standard deviation of ROP* (+) are the top 3 important firm-level predictors of insurers' failure. The former two indicators have a negative impact on the probability of firm failure based on the logit estimation, while the latter indicator has a positive impact. Mutual insurers have less profit pressure than stock firms and thus usually take fewer risks (Lamm-Tennant & Starks, 1993). *Composite* captures the line of business diversification of an insurer; more diversified insurers are less vulnerable than one-sector insurers. *Moving standard deviation of ROP* captures insurers' business volatility; firms with higher volatility are more likely to fail.

We categorize other firm-level predictors into ten groups as described in Section 2. Profitability, size, and capital adequacy are important drivers of firm failures, followed by investment and business volatility measures. These relatively important insurer failure predictors are consistent with those identified in the previous literature (see e.g., Chen & Wong, 2004; Leverty & Grace, 2010, 2012; Pasiouras & Gaganis, 2013; Zhang & Nielson, 2015).

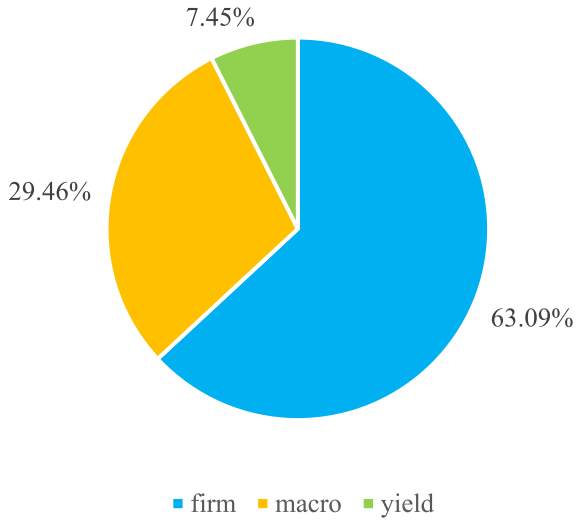


Fig. 3. Relative importance of firm, macro, and yield information in predicting firm failure.

Fig. 5 presents the relative importance of macroeconomic variables. *Change in IMF 1-year T-bill rate* (+), *GDP* (−), and *yearly return of MSCI* (−) are the top-3 important macroeconomic predictors. *Change in IMF 1-year T-bill rate* captures the capital liquidity change; a large increase in T-bill rate indicates liquidity shortage and thus a higher probability of firm failure. *GDP* is a widely used measure of economic development and market size; firms in larger and more developed economies tend to have fewer failure events. *Yearly return of MSCI* is the most widely used international equity benchmark by institutional investors (Hau et al., 2010). A higher return of MSCI indicates a better financial market environment and a booming economy, which leads to fewer firm failures.

We categorize other macroeconomic predictors into six groups, as described in Section 2. The major drivers of firm failures are economic growth and market maturity. The demographic factors are relatively less important.

Fig. 6 presents the relative importance of yield curves with different maturities. The 1-year yield to maturity of the sovereign debt (+) has the largest impact, as it reflects the financial environment in the nearest short term. A higher yield indicates scarcer liquidity and thus is positively correlated with the failure probability based on the logit estimation. The difference in predictability among other yield curves is small.

4.4. Long-term forecasting power: detecting distressed firms

The exercise above explores machine learning techniques to predict firm failure in the next year. This subsection explores the same techniques to predict whether a firm will go bankrupt during the next 2 years; we are detecting distressed firms. This exercise provides insights into the long-term forecasting capability of the model. As shown in Table 9, the SANNs are able to detect distressed firms 2 years in advance.

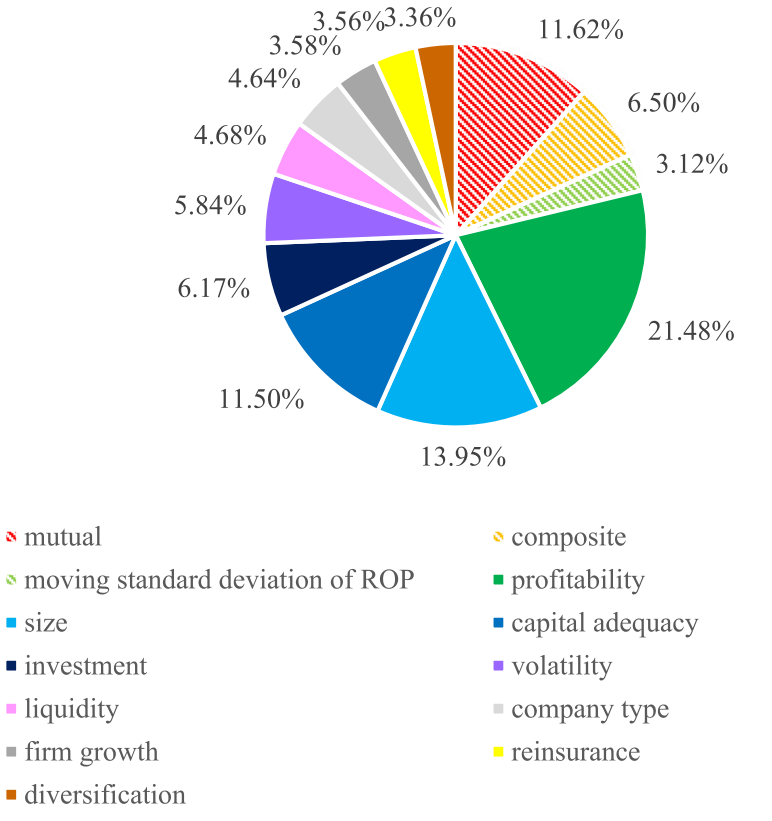


Fig. 4. Relative importance of firm characteristics in predicting firm failure.

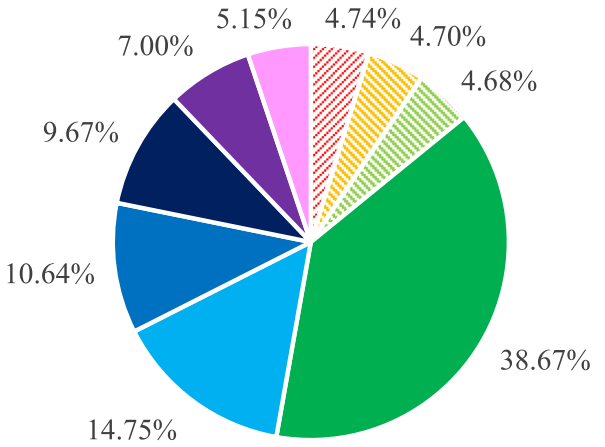


Fig. 5. Relative importance of macroeconomic indicators in predicting firm failure.

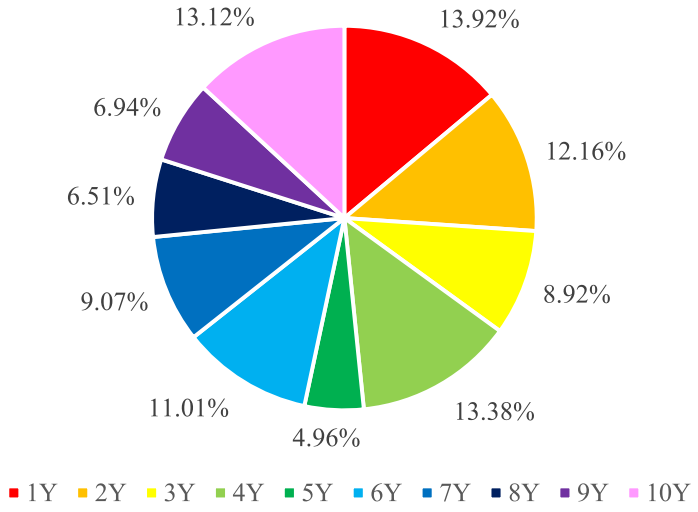


Fig. 6. Relative importance of yield curves with different maturities in predicting firm failure.

Table 9
Out-of-sample AUC in Detecting Distressed Firms.

Model	logit	RF	XGBoost	NN[4]	NN[4,2]	NN[4,3,2]	NN[8,6,4,2]	NN[8,6,4,3,2]	SANN0	SANN1
Information Explored										
Firm	0.8344	0.8348	0.8670	0.8551	0.8552	0.8508	0.8561	0.8284	0.8508	0.8508
Firm and yield	0.7943	0.7940	0.7702	0.8488	0.8140	0.8317	0.8281	0.8411	0.8505	0.8560
Firm and macro	0.8455	0.7923	0.8247	0.8494	0.8460	0.8449	0.8493	0.8216	0.8712	0.8650
All	0.8048	0.7825	0.7830	0.8443	0.8332	0.8356	0.8389	0.8028	0.8789	0.8707

Generally, NN-type models dominate the logit model. Taking the SANN0 as an example, compared with the logit model based on firm information only, firm and yield information, firm and macro information, and all information, the SANN0 increases the AUC of the logit model by 1.97%, 7.07%, 3.04%, and 9.20%, respectively.

The logit model and traditional machine learning methods again suffer from the curse of high dimensionality, unable to extract nonredundant predictive information from yield and macro information. Taking the logit (NN[4,3,2]) model based on firm information only as an example, adding yield information decreases the AUC by 4.80% (2.24%), while adding yield and macro information together decreases the AUC by 3.54% (1.79%).

In contrast, the SANNs are still able to extract information from the extra data. Taking the SANN0 as an example, compared with the SANN0 based on firm information only, adding yield information slightly decreases the AUC by 0.03%, adding macro information improves the AUC by 2.40%, and adding both types of information improves the AUC by 3.31%. This result again validates the considerations of macroeconomic and yield information in failure/insolvency prediction.

In the meantime, cross-category interaction is redundant in detecting distressed firms. The SANN1, which allows further interaction, cannot improve the AUC of the SANN0. Though the SANN1 can still handle high-dimensional predictors, the average AUC difference between these two models is 0.22%.

In sum, our findings are novel to the literature on both machine learning and predicting insolvency and failure in that we show that NNs, particularly SANNs, substantially outperform conventional models in predicting firm failure. We also show that disciplining the machine learning model instead of relying on the universal approximation ability of NNs improves the model's performance through a more satisfactory ability to explore nonredundant predictive information across different types/categories of information. In particular, further interaction between different types/categories of information is at most only marginally necessary for predicting firm failure and detecting distressed firms.

5. Robustness checks

Failures of life insurers and non-life insurers. We repeat the same firm failure prediction procedures in life insurers and non-life insurers subsamples, since in most markets they are operated by separate entities (Biener et al., 2021). We use 58 firm-level failure indicators for the life sample, and 59 firm-level indicators for the nonlife sample.[11] The macro and yield curve information remains the same as in our baseline prediction. We report the out-of-sample performance of the prediction models in Table

[11] Compared to the baseline sample, we drop the dummies *life* and *composite* in both the life and nonlife samples. 5 life-specific indicators are added in the life sample (*expense ratio*, *benefits paid/ net premiums written*, *combined ratio (life)*, *benefits paid*, and *net expenses (life)*); 6 nonlife-specific indicators are added in the nonlife sample (*loss ratio*, *operating expense ratio*, *combined ratio*, *operating ratio*, *net expenses*, and *net investment/ income*).

Table 10
Out-of-sample AUC in Detecting Failures of Life Insurers.

Model	logit	RF	XGBoost	NN[4]	NN[4,2]	NN[4,3,2]	NN[8,6,4,2]	NN[8,6,4,3,2]	SANN0	SANN1
Information Explored										
Firm	0.8426	0.8292	0.8357	0.8655	0.8886	0.8710	0.8656	0.8615	0.8710	0.8710
Firm and yield	0.8375	0.8001	0.8091	0.8530	0.8353	0.8077	0.8644	0.8183	0.8687	0.8721
Firm and macro	0.6138	0.7863	0.7661	0.8718	0.8290	0.8735	0.8323	0.8492	0.8920	0.8932
All	0.6182	0.8267	0.7607	0.8654	0.8693	0.8392	0.8526	0.8475	0.8921	0.8892

Table 11
Out-of-sample AUC in Detecting Failures of Nonlife Insurers.

Model	logit	RF	XGBoost	NN[4]	NN[4,2]	NN[4,3,2]	NN[8,6,4,2]	NN[8,6,4,3,2]	SANN0	SANN1
Information Explored										
Firm	0.8524	0.8520	0.8532	0.8629	0.8665	0.8588	0.8490	0.8643	0.8588	0.8588
Firm and yield	0.7807	0.8048	0.8212	0.8627	0.8532	0.8555	0.8517	0.8615	0.8669	0.8732
Firm and macro	0.8355	0.8677	0.7927	0.8169	0.8267	0.8332	0.8172	0.8208	0.8553	0.8705
All	0.8081	0.8526	0.7732	0.8202	0.8125	0.8184	0.8167	0.8292	0.8746	0.8732

Table 12
Out-of-sample AUC in Detecting Firm Failures (Excluding Being Merged).

Model	logit	RF	XGBoost	NN[4]	NN[4,2]	NN[4,3,2]	NN[8,6,4,2]	NN[8,6,4,3,2]	SANN0	SANN1
Information Explored										
Firm	0.8646	0.8612	0.8751	0.8795	0.8848	0.8837	0.8992	0.8453	0.8837	0.8837
Firm and yield	0.7961	0.7982	0.7875	0.8712	0.8483	0.8576	0.8661	0.8635	0.8794	0.8838
Firm and macro	0.8608	0.8285	0.8100	0.8706	0.8706	0.8650	0.8775	0.8637	0.9040	0.8996
All	0.8450	0.7968	0.8170	0.8667	0.8558	0.8503	0.8520	0.8680	0.8954	0.9035

10 for life insurers and in Table 11 for non-life insurers. The empirical patterns reconfirm our previous findings.

Alternative definition of failure events. We exclude mergers from failure events as failure events recorded by A.M. Best include both voluntary and involuntary exits. Thus, being merged does not necessarily indicate distress or insolvency of insurers (BarNiv & McDonald, 1992). Given this refined definition, the out-of-sample performance reported in Table 12 is consistent with our baseline results.

6. Conclusion

In this paper, we explore the SANN, a novel machine learning approach, to predict insurer failure by leveraging firm characteristics, macroeconomic indicators, and yield information. This innovative technology facilitates the extraction of nonredundant predictive information from different categories of predictors. Our analysis demonstrates that the SANN enhances the out-of-sample failure prediction accuracy for European insurers, outperforming both traditional machine learning models and logistic regressions. Notably, this improvement is achieved on the same dataset used in previous insurer failure predictions (Eling & Jia, 2018). Furthermore, we demonstrate that macroeconomic and yield data along with firm characteristics provide nonredundant predictive insights. This highlights the importance of incorporating environmental factors and return rates in sufficiently powerful insurer failure prediction models (Chen & Yuan, 2017; Delsing et al., 2019; Sun & Wei, 2014).

This research contributes to the literature in two dimensions. First, we demonstrate that the SANN effectively extracts nonredundant predictive information from high-dimensional data, a challenge that traditional machine learning techniques struggle to overcome. Second, for the first time, we integrate the SANN with economic priors into insurer insolvency/failure prediction. This approach significantly enhances out-of-sample and out-of-time prediction performance, leveraging information that is not captured by firm variables but is crucial for predicting failure and identifying distress.

CRediT authorship contribution statement

Canchun He: Writing – review & editing, Resources, Data curation, Formal analysis. **Dejin Huang:** Validation, Software, Formal analysis, Data curation. **Ruo Jia:** Writing – review & editing, Validation. **Xi Wang:** Writing – original draft, Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xi Wang reports financial support was provided by Peking University. Xi Wang reports a relationship with Peking University that includes: employment and funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors acknowledge the financial support from the Natural Science Foundation of China (No. 72595832,

Table A.1
The Standard Deviations of AUC.

Model	NN[4]	NN[4,2]	NN[4,3,2]	NN[8,6,4,2]	NN[8,6,4,3,2]	SANN0	SANN1
Information Explored							
Firm	0.000759	0.00079	0.001293	0.001133	0.002287	0.001293	0.001293
Firm and yield	0.000883	0.002213	0.00184	0.00214	0.002426	0.002895	0.00254
Firm and macro	0.001061	0.001579	0.002052	0.004371	0.001745	0.002147	0.00274
All	0.000643	0.003106	0.002909	0.000746	0.001355	0.002179	0.001419

Notes: This table shows the robustness of the ensembled 10 neural networks described in Section 3.2. Within the 10 neural networks, we perform 10 rounds of sampling, each time selecting 9 neural networks, and each neural network is left out once in one of the rounds. After each round of sampling, the average predicted value of the selected 9 models is used as the final prediction, and the AUC is calculated. Thus, for each neural network model under each information set, we obtain 10 AUC values. In the table, we report the standard deviations of the 10 AUCs.

Table A.2
Out-of-sample AUC in Detecting Firm Failures without Cost-sensitive Learning.

Model	logit	RF	XGBoost	NN[4]	NN[4,2]	NN[4,3,2]	NN[8,6,4,2]	NN[8,6,4,3,2]	SANN0	SANN1
Information Explored										
Firm	0.8496	0.8306	0.8642	0.8616	0.8604	0.8606	0.8688	0.8600	0.8606	0.8606
Firm and yield	0.7956	0.8309	0.8700	0.8509	0.8633	0.8538	0.8652	0.8607	0.8614	0.8529
Firm and macro	0.8114	0.8211	0.8282	0.8358	0.8297	0.8363	0.8364	0.8280	0.8650	0.8705
All	0.8106	0.8156	0.8180	0.8437	0.8407	0.8363	0.8281	0.8253	0.8667	0.8719

Notes: This table reports the out-of-sample AUC of various prediction models described in Section 3.2. Different with the analysis within the main text, the models here are all trained without "cost-sensitive learning" sample.

Table A.3
Hyperparameters for Model Tuning.

RF	XGBoost	ANN/SANN
#Trees ∈ {10,20,30,40,50,60,70,80,90}; Max_depth ∈ {3,4,5,6}; Max_features ∈ {3,4,5,6}.	#Trees ∈ {10,30,50,70,90}; Max_depth ∈ {3,4,5,6}; Reg_lambda ∈ {0.001,0.01,0.1,1,10}.	Learning Rate = 0.001; Batch Size = 128; Patience = 50; Ensemble = 10; Dropout Rate = 0.1;

Note: The table describes the hyperparameters that we tune in each machine learning method.

72173005), the Social Science Foundation of China (23BJL025, 23&ZD178), and School of Economics, Peking University.

Appendix A. More results

This appendix reports the empirical results mentioned in the main text. Table A.1 reports the standard deviations of AUC. To compute this standard deviation for a given dataset, we perform 10 times model sampling, within which we select nine models out of ten and leverage their ensembled output for prediction, for which we can compute the AUC. With this process, we can compute standard deviation of AUC for each prediction model. As shown in Table A.1, the standard deviations of AUC for each model are all very small and less than 0.01. Therefore, as explained in footnote 9 of the main text, the ensemble learning strategy that we have adopted in this work is robust.

Moreover, Table A.2 below reports the empirical analysis without "cost-sensitive learning". Whether resampling the raw data to keep sample balance is the only difference between the models explored in our main text (e.g., Table 7) and the models investigated in Table A.2. As shown in Table A.2, our main results remain unchanged: only the SANN can effectively extract nonredundant predictive information from high-dimensional data, which is a challenge that traditional machine learning techniques struggle to overcome.

Finally, Table A.3 reports the set of potential hyperparameters used for tuning each machine learning method during the model training. As explained in the main text, we follow the approach in the machine learning and asset pricing literature and select tuning parameters adaptively from the data in a validation sample.

Data and code availability

Except for the proprietary insurer-level data obtained from A.M. Best, all data and code necessary to reproduce the results in this study are available at: https://github.com/CanchunHe/Machine-Learning-and-Insurer-Failure-Prediction. The A.M. Best data used in this study may be obtained from A.M. Best ("Best's Insurance Reports - L/NL, Non US"). Access to these data is subject to A.M. Best's licensing terms, and the cost may vary depending on the replicator's institution type and country. To the extent permitted by the licensing agreement, the authors will assist replicators in obtaining the data for replication purposes.

References

Ang, A., & Piazzesi, M. (2003). A no-arbitrage vector autoregression of

term structure dynamics with macroeconomic and latent variables. *Journal of Monetary Economics*, *50*(4), 745–787.
Bai, J., & Ng, S. (2008). Forecasting economic time series using targeted predictors. *Journal of Econometrics*, *146*(2), 304–317.
Barboza, F., Kimura, H., & Altman, E. (2017). Machine learning models and bankruptcy prediction. *Expert Systems with Applications*, *83*, 405–417.
BarNiv, R., & McDonald, J. B. (1992). Identifying financial distress in the insurance industry: A synthesis of methodological and empirical issues. *Journal of Risk and Insurance*, *54*, 3–573.
Bauer, J., & Agarwal, V. (2014). Are hazard models superior to traditional bankruptcy prediction approaches? A comprehensive test. *Journal of Banking and Finance*, *40*, 432–442.
Best, A. M. (2002-2013). Best's insurance reports, databases version 2007 2010, and 2013.
Best, A. M. (2010, 2013). Best's corporate changes and retirements. http://www.ambest.com/ccr (2013).
Bianchi, D., Büchner, M., & Tamoni, A. (2021). Bond risk premiums with machine learning. *Review of Financial Studies*, *34*(2), 1046–1089.
Biener, C., Eling, M., & Jia, R. (2017). The structure of the global reinsurance market: An analysis of efficiency, scale, and scope. *Journal of Banking and Finance*, *77*, 213–229.
Biener, C., Eling, M., & Jia, R. (2021). Globalisation: blessing or curse? Evidence from the insurance industry. *European Journal of International Management*, *15*(2–3), 457–483.
Borchert, P., Coussement, K., De Caigny, A., & De Weerdt, J. (2023). Extending business failure prediction models with textual website content using deep learning. *European Journal of Operational Research*, *306*(1), 348–357.
Brockett, P. L., Cooper, W. W., Golden, L. L., & Pitaktong, U. (1994). A neural network method for obtaining an early warning of insurer insolvency. *Journal of Risk and Insurance*, *40*, 2–424.
Brockett, P. L., Cooper, W. W., Golden, L. L., & Xia, X. (1997). A case study in applying neural networks to predicting insolvency for property and casualty insurers. *Journal of the Operational Research Society*, *48*, 1153–1162.
Brockett, P. L., Golden, L. L., Jang, J., & Yang, C. (2006). A comparison of neural network, statistical methods, and variable choice for life insurers' financial distress prediction. *Journal of Risk and Insurance*, *73*(3), 397–419.
Browne, M. J., & Hoyt, R. E. (1995). Economic and market predictors of insolvencies in the property-liability insurance industry. *Journal of Risk and Insurance*, *30*, 9–327.
Campbell, J. Y., Hilscher, J., & Szilagyi, J. (2008). In search of distress risk. *Journal of Finance*, *63*(6), 2899–2939.
Carmona, P., Climent, F., & Momparler, A. (2019). Predicting failure in the US banking sector: An extreme gradient boosting approach. *International Review of Economics Finance*, *61*, 304–323.
Chan, W. S., Yang, H., & Zhang, L. (2003). Some results on ruin probabilities in a two-dimensional risk model. *Insurance: Mathematics & Economics*, *32*(3), 345–358.
Chava, S., & Jarrow, R. A. (2004). Bankruptcy prediction with industry effects. *Review of Finance*, *8*(4), 537–569.
Chen, L., Pelger, M., & Zhu, J. (2023). Deep learning in asset pricing. Management science.
Chen, W. H., & Shih, J. Y. (2006). A study of Taiwan's issuer credit rating systems using support vector machines. *Expert Systems with Applications*, *30*(3), 427–435.
Chen, R., & Wong, K. A. (2004). The determinants of financial health of Asian insurance companies. *Journal of Risk and Insurance*, *71*(3), 469–499.
Chen, Y., & Yuan, Z. (2017). A revisit to ruin probabilities in the presence of heavy-tailed insurance and financial risks. *Insurance: Mathematics & Economics*, *73*, 75–81.
Cheng, J., & Weiss, M. A. (2012). The role of RBC, hurricane exposure, bond portfolio duration, and macroeconomic and industry-wide factors in property–liability insolvency prediction. *Journal of Risk and Insurance*, *79*(3), 723–750.
Climent, F., Momparler, A., & Carmona, P. (2019). Anticipating bank distress in the Eurozone: An extreme gradient boosting approach. *Journal of Business Research*, *101*, 885–896.
Cole, R., & Wu, Q. (2017). Hazard versus probit in predicting U.S. bank failures: A regulatory perspective over two crises.
Cox, R. A., & Wang, G. W. Y. (2014). Predicting the US bank failure: A discriminant analysis. *Economic Analysis and Policy*, *44*(2), 202–211.
De Caigny, A., Coussement, K., & De Bock, K. W. (2018). A new hybrid classification algorithm for customer churn prediction based on logistic regression and decision trees. *European Journal of Operational Research*, *269*, 760–772.
Delsing, G. A., Mandjes, M. R. H., Spreij, P. J. C., & Winands, E. M. M. (2019). An optimization approach to adaptive multi-dimensional capital management. *Insurance: Mathematics & Economics*, *84*, 87–97.
Diebold, F. X., Rudebusch, G. D., & Aruoba, S. B. (2006). The macroeconomy and the yield curve: a dynamic latent factor approach. *Journal of Econometrics*, *131*(1–2), 309–338.
Doumpos, M., Andriosopoulos, K., Galariotis, E., Makridou, G., & Zopounidis, C. (2017). Corporate failure prediction in the European energy sector: A multicriteria approach and the effect of country characteristics. *European Journal of Operational Research*, *262*(1), 347–360.
Eelbode, T., Bertels, J., Berman, M., Vandermeulen, D., Maes, F., Bisschops, R., & Blaschko, M. B. (2020). Optimization for medical image segmentation: theory and practice when evaluating with dice score or jaccard index. *IEEE Transactions on Medical Imaging*, *39*(11), 3679–3690.
Eling, M., & Jia, R. (2018). Business failure, efficiency, and volatility: Evidence from the European insurance industry. *International Review of Financial Analysis*, *59*, 58–76.
Feng, R. (2009). On the total operating costs up to default in a renewal risk model. *Insurance: Mathematics & Economics*, *45*(2), 305–314.
Feng, R. (2011). An operator-based approach to the analysis of ruin-related quantities in jump diffusion risk models. *Insurance: Mathematics & Economics*, *48*(2), 304–313.
Forgione, A. F., & Migliardo, C. (2018). Forecasting distress in cooperative banks: The role of asset quality. *International Journal of Forecasting*, *34*(4), 678–695.
Funahashi, K. I. (1989). On the approximate realization of continuous mappings by neural networks. *Neural Networks*, *2*(3), 183–192.
Ge, S. (2022). How do financial constraints affect product pricing? Evidence from weather and life insurance premiums. *Journal of Finance*, *77*(1), 449–503.
Geng, R., Bose, I., & Chen, X. (2015). Prediction of financial distress: An empirical study of listed Chinese companies using data mining. *European Journal of Operational Research*, *241*(1), 236–247.
Gogas, P., Papadimitriou, T., & Agrapetidou, A. (2018). Forecasting bank failures and stress testing: A machine learning approach. *International Journal of Forecasting*, *34*(3), 440–455.
Green, J., Hand, J. R., & Zhang, X. F. (2017). The characteristics that provide independent information about average US monthly stock returns. *Review of Financial Studies*, *30*(12), 4389–4436.
Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, *33*(5), 2223–2273.
Harrington, S. E. (2009). The financial crisis, systemic risk, and the future of insurance regulation. *Journal of Risk and Insurance*, *76*(4), 785–819.
Hau, H., Massa, M., & Peress, J. (2010). Do demand curves for currencies slope down? Evidence from the MSCI global index change. *Review of Financial Studies*, *23*(4), 1681–1717.
Ho, Y., & Wookey, S. (2019). The real-world-weight cross-entropy loss function: Modeling the costs of mislabeling. *IEEE Access*, *8*, 4806–4813.
Hornik, K., Stinchcombe, M., & White, H. (1989). Multilayer feedforward networks are universal approximators. *Neural Networks*, *2*(5), 359–366.
du Jardin, P. (2021). Forecasting corporate failure using ensemble of self-organizing neural networks. *European Journal of Operational Research*, *288*(3), 869–885.
Kočišová, K., & Mišanková, M. (2014). Discriminant analysis as a tool for forecasting company's financial health. *Procedia-Social and Behavioral Sciences*, *110*, 1148–1157.
Koijen, R. S., & Yogo, M. (2022). The fragility of market risk insurance. *Journal of Finance*, *77*(2), 815–862.
Konstantinides, D. G., & Li, J. (2016). Asymptotic ruin probabilities for a multidimensional renewal risk model with multivariate regularly varying claims. *Insurance: Mathematics & Economics*, *69*, 38–44.
Kumar, P. R., & Ravi, V. (2007). Bankruptcy prediction in banks and firms via statistical and intelligent techniques–A review. *European Journal of Operational Research*, *180*(1), 1–28.

Lamm-Tennant, J., & Starks, L. T. (1993). Stock versus mutual ownership structures: The risk implications. *Journal of Business*, *2*, 9–46.
Lee, S. H., & Urrutia, J. L. (1996). Analysis and prediction of insolvency in the property-liability insurance industry: A comparison of logit and hazard models. *Journal of Risk and Insurance*, *12*, 1–130.
Leverty, J. T., & Grace, M. F. (2010). The robustness of output measures in property-liability insurance efficiency studies. *Journal of Banking and Finance*, *34*(7), 1510–1524.
Leverty, J. T., & Grace, M. F. (2012). Dupes or incompetents? An examination of management's impact on firm distress. *Journal of Risk and Insurance*, *79*(3), 751–783.
Martin, D. (1977). Early warning of bank failure: A logit regression approach. *Journal of Banking and Finance*, *1*(3), 249–276.
Masters, T. (1993). *Practical neural network recipes in c++*. New York: Academic Press.
McDonald, R., & Paulson, A. (2015). AIG in hindsight. *Journal of Economic Perspectives*, *29*(2), 81–106.
Messai, A. S., & Gallali, M. I. (2015). Financial leading indicators of banking distress: A micro prudential approach-evidence from europe. *Asian Social Science*, *11*(21), 78.
Meyer, P. A., & Pifer, H. W. (1970). Prediction of bank failures. *Journal of Finance*, *25*(4), 853–868.
Mselmi, N., Lahiani, A., & Hamza, T. (2017). Financial distress prediction: The case of french small and medium-sized firms. *International Review of Financial Analysis*, *50*, 67–80.
Ohlson, J. A. (1980). Financial ratios and the probabilistic prediction of bankruptcy. *Journal of Accounting Research*, 109–131.
Pasiouras, F., & Gaganis, C. (2013). Regulations and soundness of insurance firms: International evidence. *Journal of Business Research*, *66*(5), 632–642.
Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, *12*, 2825–2830.
Petropoulos, A., Siakoulis, V., Stavroulakis, E., & Vlachogiannakis, N. E. (2020). Predicting bank insolvencies using machine learning techniques. *International Journal of Forecasting*, *36*(3), 1092–1113.
Rauch, J., & Wende, S. (2015). Solvency prediction for property-liability insurance companies: Evidence from the financial crisis. *Geneva Papers on Risk and Insurance-Issues and Practice*, *40*, 47–65.
Sinkey, J. F. (1975). A multivariate statistical analysis of the characteristics of problem banks. *Journal of Finance*, *30*(1), 21–36.
Sun, Y., & Wei, L. (2014). The finite-time ruin probability with heavy-tailed and dependent insurance and financial risks. *Insurance: Mathematics & Economics*, *59*, 178–183.
Tam, K. Y., & Kiang, M. Y. (1992). Managerial applications of neural networks: the case of bank failure predictions. *Management Science*, *38*(7), 926–947.
Tanaka, K., Kinkyo, T., & Hamori, S. (2016). Random forests-based early warning system for bank failures. *Economics Letters*, *148*, 118–121.
Tian, S., Yu, Y., & Guo, H. (2015). Variable selection and corporate bankruptcy forecasts. *Journal of Banking and Finance*, *52*, 89–100.
Ting, K. M. (2002). An instance-weighting method to induce cost-sensitive trees. *IEEE Transactions on Knowledge and Data Engineering*, *14*(3), 659–665.
Veganzones, D., Séverin, E., & Chlibi, S. (2023). Influence of earnings management on forecasting corporate failure. *International Journal of Forecasting*, *39*(1), 123–143.
Wang, G., Ma, J., & Yang, S. (2014). An improved boosting based on feature selection for corporate bankruptcy prediction. *Expert Systems with Applications*, *41*(5), 2353–2361.
Wang, G., Wong, K. W., & Lu, J. (2020). AUC-based extreme learning machines for supervised and semi-supervised imbalanced classification. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, *51*(12), 7919–7930.
Zhang, L., & Nielson, N. (2015). Solvency analysis and prediction in property–casualty insurance: Incorporating economic and market predictors. *Journal of Risk and Insurance*, *82*(1), 97–124.